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How Viscosity-Based and Density-Based AICDs Respond Differently to Annulus Separation and Cascade Closing

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Abstract

For almost two decades, autonomous inflow control devices (AICDs) have proven extremely successful in preventing excessive inflow of unwanted gas and water from oil reservoirs. Two fundamentally different autonomous technologies exist in the market: viscosity-based and density-based, where the latter is defined by being completely insensitive to viscosity and Reynolds number. With multiple autonomous devices in each well compartment, it is critical to understand how they interact and respond collectively to fluid separation in the annulus.

The analysis is limited to a single well compartment that produces ultra-light oil and water only, but the findings are also relevant when choking back gas in oil wells. For this compartment, a direct comparison is made between eight viscosity-based devices and two density-based devices, which have the same total flow capacity when all devices are initially open for single-phase oil. A state-of-the-art modelling approach is used to predict the fluid separation for increasing water cuts, and to evaluate the risk of so-called "cascade closing"—a phenomenon in which the closure of the lowest device causes the water level to rise and trigger the closure of the next device above, continuing upward in a cascading fashion.

It is shown how viscosity-based and density-based devices exhibit a fundamentally different collective response to a gradually increasing zonal water cut. This can be partly explained by sensitivity vs. insensitivity to flow rate, and partly by non-binary (gradual restriction) vs. binary (on/off) behaviour. The differences between the two technologies are more pronounced when the pressure drop across the lower completion increases. The modelling also shows that when the selected density-based technology has a customizable closing threshold, the choice of water fraction at which the device "closes" can have a significant impact.

Unintended cascade closing may potentially result in unpredictable behaviour and excessive choking of the inflow at low and intermediate zonal water cuts. When selecting autonomous inflow control technology and customizing its performance for any given well, it is therefore strongly recommended to simulate this effect at the zonal level and subsequently upscale the results for use in reservoir simulations.

Introduction

Since the first implementation of AICDs at the Troll field in 2008 ([Halvorsen et al. 2012](#)), such technologies have become gradually more popular. The first AICD technologies were viscosity-based and therefore required a viscosity contrast to distinguish between desired and undesired fluids. In the beginning, they were primarily used to choke back free gas in oil wells, where the viscosity contrast is always significant. Then, as the viscosity-based technologies have evolved, they have also become capable of successfully choking back formation water ([Ali et al. 2023](#)).

Below a certain oil viscosity, however, the oil/water viscosity contrast becomes too small to achieve a meaningful operational envelope that provides high choking efficiency for water and sufficient robustness against reservoir uncertainties as well as robustness against the pressure and rate development during the life of the well. This has been the main motivation for the development of density-based AICDs, which have a fundamentally different working principle, flow performance and operational envelope ([Killie et al. 2021](#); [Mallawany et al. 2023](#)). The main differences between viscosity-based and density-based AICD technologies will be summarized in the following section before their performance will be analysed in more detail for a constructed case study.

Viscosity-Based vs. Density-Based AICDs

A detailed comparison between viscosity-based and density-based AICD technologies has already been provided in a previous SPE-paper ([Killie and Grüner 2024](#)). Only the most relevant differences will therefore be repeated in this section.

Viscosity-based technologies are sometimes referred to as being "viscosity- and density-dependent" because their working principle does not depend on viscosity alone, but rather on the Reynolds number, which contains both viscosity and density. Furthermore, there may be an additional pressure loss through a restriction or nozzle, which depends on density. This should however not be confused with density-based AICD technologies, which depend on density alone without any influence from viscosity, or indeed the Reynolds number.

For viscosity-based technologies, the oil-flow capacity and achievable water-choking efficiency decrease with decreasing oil/water viscosity contrast. Furthermore, they typically start choking the flow at low or medium zonal water fraction and then choke gradually more as the water fraction increases. This is different for density-based technologies, which exhibit a binary performance, meaning that they are either open with flow through a large nozzle, or "closed" (i.e. choked) with flow through a small nozzle. They close when the water fraction exceeds a certain closing criterion, which can be set to a high value, and they reopen when it drops below a certain reopening criterion. Both closing and reopening are fully autonomous and happen on-the-fly while the well is producing.

The Reynolds number depends not only on viscosity and density, but also on velocity. For a viscosity-based technology that relies on Reynolds number, a high local inflow velocity of oil will cause a high Reynolds number, but a high Reynolds number can also be caused by unwanted fluid with low viscosity. This ambiguity may explain why some viscosity-based technologies tend to have "hockey stick"-shaped performance curves with excessive choking above a certain oil rate. For density-based technologies, this effect does not exist, and the performance curves are always quadratic, both in open and closed position.

Case Study

Background

In the above-mentioned SPE-paper ([Killie and Grüner 2024](#)), a comparison was made between viscosity-based and density-based technologies for one single compartment of a well producing a medium-light oil with a viscosity of 5.6 cP. It was shown how one density-based device could provide the same flow capacity

as four viscosity-based devices. Fluid properties and performance of viscosity-based devices were based on Langaas et al. (2023), whereas the performance of the density-based device was based on Killie et al. (2021).

A similar comparison is made in the following, but this time it is based upon an example from a more recent SPE-paper (Langaas et al. 2024), which involves two important changes from the previous comparison (Killie and Grüner 2024):

- The zone is more productive and requires eight instead of four viscosity-based devices.
- The oil is not medium-light, but ultra-light with a viscosity of 0.82 cP.

Langaas et al. (2024) presented both experimental performance curves for a single viscosity-based device, a proposed set of parameters for use in the RCP function, and a powerful zonal upscaling technique based on a state-of-the-art commercial software, which accurately captures the effect of fluid separation in the annulus and the potential for cascade closing. With zonally upscaled input from this software, subsequent reservoir simulations then demonstrated how the viscosity-based technology could give both higher cumulative oil production and significantly lower water and gas production. In the following, it will be shown how a density-based technology would perform under similar conditions.

Modelling Assumptions

As a starting point, only a single zone is considered, which is 100 m long and has 1° inclination angle from horizontal, such that water in the annulus starts to accumulate at the lower end. Gas is neglected, and the properties of oil and water are the same as in Langaas et al. (2024), i.e.

- Oil density = 757 kg/m³
- Water density = 1007 kg/m³
- Oil viscosity = 0.82 cP
- Water viscosity = 0.46 cP

In Langaas et al. (2024), this zone was populated with eight viscosity-based devices, i.e. one for every 12.5-m joint, as illustrated in Fig. 1. Measured performance data for each individual device are shown in Fig. 2 for oil and water along with theoretical performance curves based on the parameters in Table 1 recommended for the extended RCP function:

$$\Delta p = \left(\frac{\rho_{mix}}{\rho_{cal}} \right)^z \left(\frac{\mu_{cal}}{\mu_{mix}} \right)^y \rho_{mix} s \left(\frac{q}{q_{cal}} \right)^x, \quad (1)$$

where

$$\rho_{mix} = \alpha_{oil}^a \rho_{oil} + \alpha_{wat}^b \rho_{wat} + \alpha_{gas}^c \rho_{gas}, \quad (2)$$

$$\mu_{mix} = \alpha_{oil}^d \mu_{oil} + \alpha_{wat}^e \mu_{wat} + \alpha_{gas}^f \mu_{gas}. \quad (3)$$

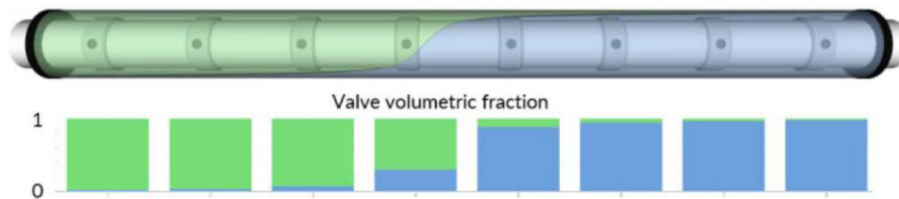


Figure 1—Single zone with eight viscosity-based AICDs, copied from Langaas et al. (2024).

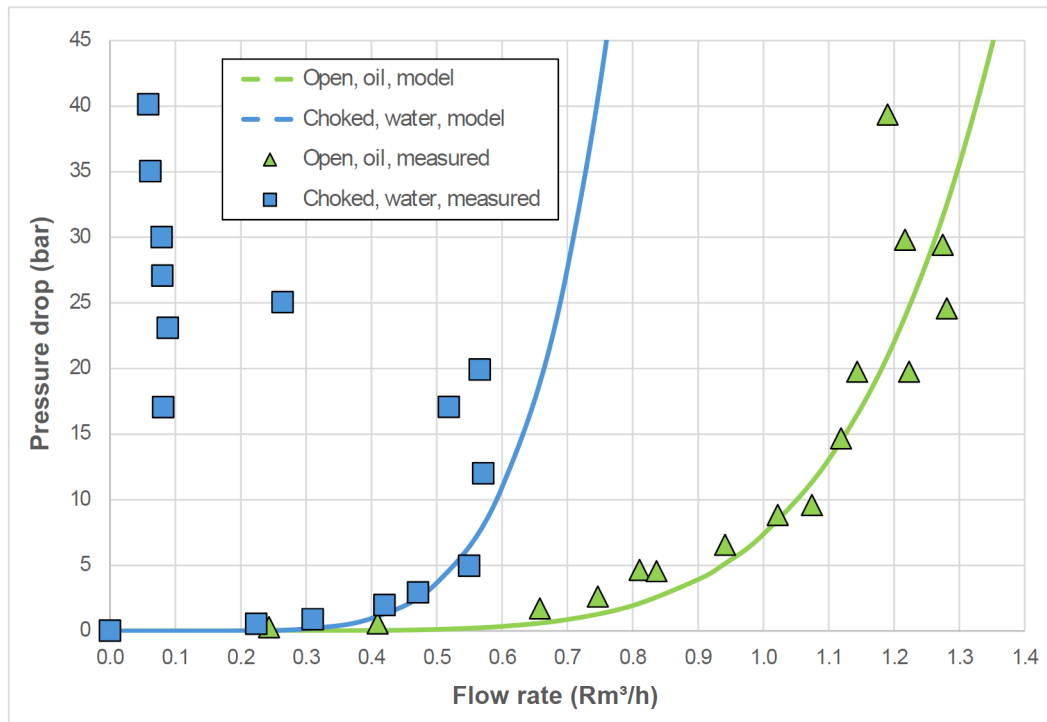


Figure 2—Measured data and theoretical performance curves for a viscosity-based AICD, based on Langaas et al. (2024).

Table 1—Parameters for extended RCP function.

ρ_{cal}	μ_{cal}	q_{cal}	x	y	s	z	a	b	c	d	e	f
kg/m ³	cP	m ³ /d	-	-	bar·m ³ /kg	-	-	-	-	-	-	-
1000	1	1	6	5	2.5E-11	1	1.5	5	1	1	1	1

In general, density-based technologies have much higher flow capacity in the open position than viscosity-based technologies—especially at low oil viscosities. For the zone described above, only two density-based devices should therefore be sufficient. This is illustrated in Fig. 3, where the solid curves represent the combined flow capacity for oil and water, respectively, through two density-based devices, more specifically two DAR[®] Assemblies, each with a Water Control Module (WCM). The nozzle diameter of each device is $d_{open} = 6.1$ mm in open position and $d_{closed} = 2.5$ mm in closed position. When switching from open to closed position at a constant pressure drop, the volumetric reduction in flow rate—or choking efficiency E —is then 85%, which is insensitive to pressure drop:

$$E \equiv \frac{q_{oil} - q_{wat}}{q_{oil}} = 1 - \sqrt{\frac{\rho_{oil}}{\rho_{wat}}} \left(\frac{d_{closed}}{d_{open}} \right)^2 = 85\%. \quad (4)$$

The dashed curves in Fig. 3 are based on the parameters in Table 1, but the flow rate is multiplied by eight to represent the sum of flow from eight devices. This approach is expected to be fairly accurate for single-phase oil and water, respectively, but cannot be justified for two-phase flow due to the annulus separation effect that will be discussed in the following section.

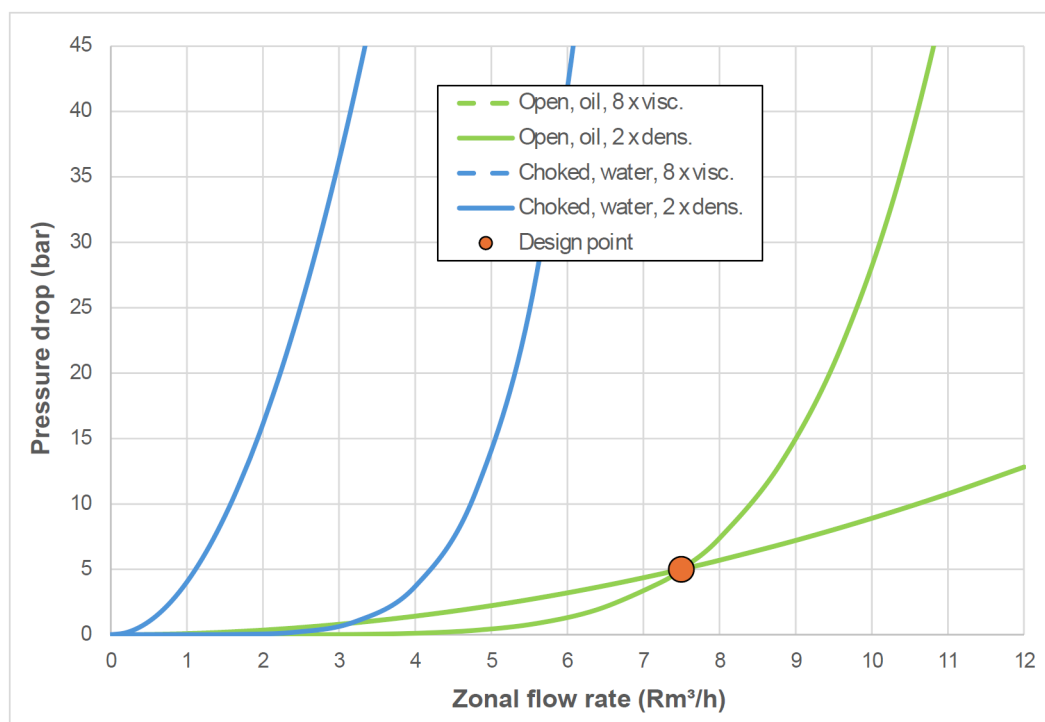


Figure 3—Zonal flow performance curves for single-phase oil and water.

For the density-based technology, the nozzle sizes, d_{open} and d_{closed} , can be made smaller and larger than the values used in Fig. 3, which would have altered the performance curves. In open position, the value 6.1 mm was selected because it gave 5 bar pressure drop for the same zonal oil rate as the sum of the eight viscosity-based devices. A pressure drop of 5 bar was mentioned as an example in Langaas et al. (2024) and is therefore defined as the design point in the following. In Fig. 3, this design point is marked where the green curves intersect.

Annulus Separation and Cascade Closing

Fluid separation in the annulus outside standalone sand screens was described by Aakre et al. (2014). Potentially negative effects of cascade closing was first explained by Rinde et al. (2017). These two interconnected phenomena have since been described in several papers (Bertram et al. 2021; Malbrel et al. 2024; Langaas et al. 2025; Brekke et al. 2025). A detailed step-by-step explanation was provided by Killie and Grüner (2024).

Briefly explained, the velocities in the annulus of a well compartment are normally so low that fluids will separate. Typically, the well trajectory is not perfectly horizontal, and water will therefore start to accumulate at the lower end until it reaches the lowermost AICD. If the water fraction through that AICD is high enough to make it choke the flow, and if the water-choking efficiency is high enough to prevent all the incoming water from entering through that choked AICD, then the water in the annulus will continue to accumulate until it reaches the next AICD. This may cause the second AICD to choke the flow, such that the water continues to the third AICD, etc.

This cascade closing effect can be somewhat non-intuitive. If the well compartment produces 50% water fraction, one would intuitively assume that half of the AICDs will see oil and the other half will see water, such that only the latter half will close. However, due to the lower flow capacity for water than oil (the inherent purpose of AICDs), simple mass conservation requires that the water must distribute itself among more AICDs, such that more than half of them end up closing.

If the water-choking efficiency of each individual AICD is high, the fraction of AICDs that close in a given zone can be much higher than what the zonal water fraction would intuitively correspond to. Unless

properly accounted for in the design, the cascade closing effect may lead to excessive choking already at low-to-intermediate zonal water fractions. At higher zonal water fractions, all devices might end up choking back the flow even if the zonal water fraction is still low enough for this to be undesirable. It is therefore of paramount importance to fully understand how multiple AICDs in a zone respond collectively to a gradually increasing zonal water fraction, and to incorporate this understanding in the design process. The workflow proposed by Langaas et al. (2024) ensures this.

Gradual Choking, Belly Effect and Binarity

At individual AICD level, the two-phase response to a gradually increasing water fraction can vary significantly for different technologies, pressure drops, fluid properties and well customizations. In general, the viscosity-based technologies tend to exhibit gradual choking whereas the density-based technologies exhibit binary choking.

For the viscosity-based technologies, the choking may increase monotonously with increasing water fraction at constant pressure drop, as illustrated by the dark blue curve in Fig. 4, or it exhibits a "belly shape", as illustrated by the light blue curve. This belly shape was observed in the measured two-phase data presented by Langaas et al. (2024), and it is also captured by the parameter set in Table 1 above. For illustration purposes, all the curves in Fig. 4 are plotted with the same choking efficiency for 100% water, which is arbitrarily chosen to be 70%.

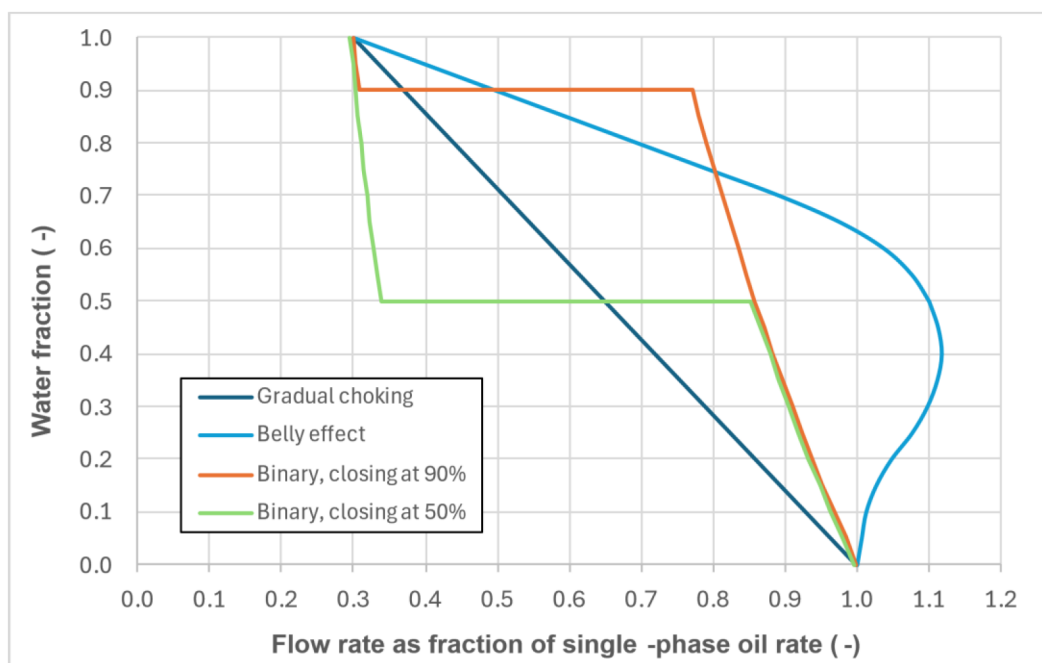


Figure 4—Illustration of different two-phase behaviours at constant pressure drop.

If the number of devices in a zone is large, the exact two-phase behaviour may be of limited importance because there will often be only one device with two-phase flow. All the other devices will predominantly see either oil or water, as in the example with eight devices in Fig. 1. Even if the devices are randomly distributed around the basepipe's circumference, an inclination angle of 1° from horizontal is sufficient to give an elevation change of 22 cm along a 12.5-meter joint, thereby promoting water accumulation that begins at the zone's lower end and leaves most devices in single-phase surroundings. This finding is supported by sensitivity simulations in the above-mentioned state-of-the-art commercial software.

Also for density-based technologies, there will typically be only one device per zone that sees two-phase flow at any point in time. However, as fewer devices are needed than for viscosity-based technologies, the

relative importance of the device with two-phase flow increases. Fig. 4 illustrates the binary behaviour of a density-based device. For the DAR Technology, it can be customized to close at any water fraction between 20% and 90%, which is insensitive to flow rate.

Going back to the case study above, the design objective should be to avoid that the last of the two density-based devices closes before the zonal water fraction reaches the level where maximum choking is desirable. This is illustrated in Fig. 5 for three different choking efficiencies, where the effective nozzle diameter in closed position is 2.0, 2.5 and 3.0 mm, respectively. If a low choking efficiency is selected (green curve for 3.0 mm nozzle), and if each individual device is configured to close at 50% in-situ water fraction, then the second device will close when the zonal water fraction reaches 60%. With higher choking efficiency (blue curve for 2.0 mm nozzle), the second device will close at 55% zonal water fraction.

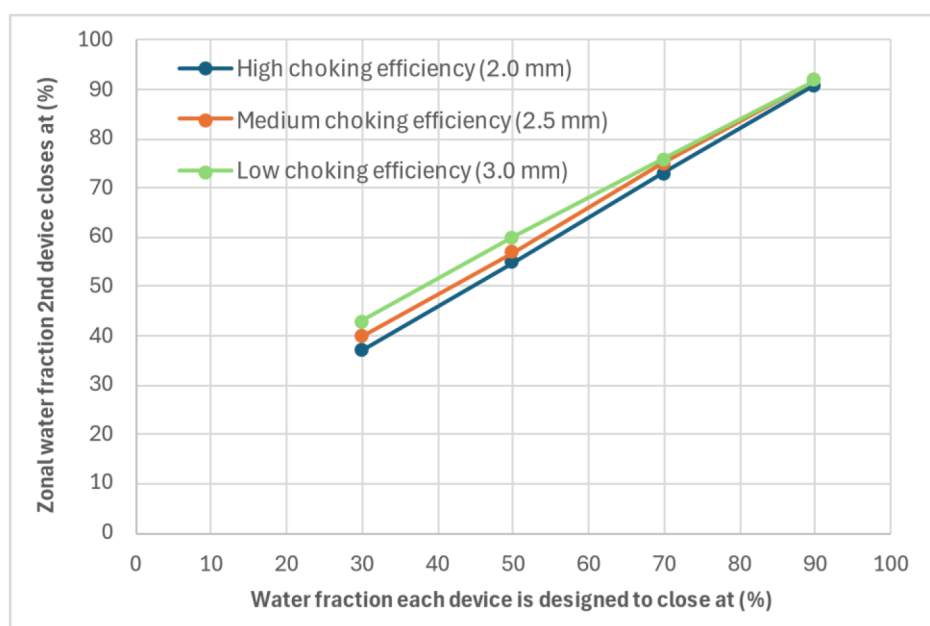


Figure 5—Zonal water fraction that the second density-based device closes at.

It can be seen how the curves in Fig. 5 converge as the water fraction increases. This means that if each density-based device is configured to close at a high in-situ water fraction of say 90%, then the second device will close at a zonal water fraction that is only marginally higher than 90%, regardless of choking efficiency.

The reason why closing of the last device is more important than closing of the first, is that the relative reduction in total zonal flow area is much larger when the last device closes. With $d_{open} = 6.1$ mm and $d_{closed} = 2.5$ mm, the flow area decreases by 42% when the first device closes and 71% when the last device closes (83% compared to the initial area), as shown in Table 2, which also includes a case with $d_{open} = 8.0$ mm and $d_{closed} = 2.0$ mm, i.e. higher choking efficiency. This observation is also relevant if the zone contains more than two density-based devices.

Table 2—Relative reduction in flow area when first and second device closes.

	$d_{open} = 6.1$ mm $d_{closed} = 2.5$ mm	$d_{open} = 8.0$ mm $d_{closed} = 2.0$ mm
Relative area reduction when 1 st device closes	42%	47%
Relative area reduction when 2 nd device closes	71%	88%
Total relative area reduction when 2 nd device closes	83%	94%

Increasing Pressure Drop Across Completion

Reasons for Increased Pressure Drop

Decreasing reservoir pressure may result in reduced pressure drop across the lower completion over time, but the pressure drop may also increase—at least temporarily—for two reasons:

- Increased annulus pressure as AICDs start to close
- Reduced bottomhole pressure if the well is produced harder

These effects will be described separately in the following.

Increased Annulus Pressure

This effect was described by [Killie and Grüner \(2024\)](#). As more and more AICDs in a zone close, the effective flow area becomes smaller. In the theoretical case that the zonal productivity index is infinitely high, the annulus pressure is constant and equal to the reservoir pressure. The pressure drop across the AICDs will then remain constant, and the flow rate is directly proportional to the total available flow area through the AICDs, at least for the density-based devices.

In practice, however, the zonal productivity index is not infinitely high. As the total available flow area and zonal flow rate decrease, the reservoir drawdown will also decrease, which means higher annulus pressure and therefore higher pressure drop across the completion in that particular zone. Lower zonal productivity index means higher pressure drop. It is therefore relevant to know how this will affect the AICDs' collective response.

Reduced Bottomhole Pressure

The productivity-index effect described above is a local, zonal effect. A reduction in bottomhole pressure—on the other hand—is a global effect that increases the pressure drop across the completion in *all* zones.

When one or more zones are gradually watered out and then choked back by the AICDs, then the well's total water cut will decrease and make the fluid column lighter in the vertical section of the well compared to a situation without AICDs. This means lower bottomhole pressure for the same tubinghead pressure. This effect is important in retrofit applications for wells that already struggle with high water holdup and limited lift capability.

The situation is different in new, prolific wells that have just started to cut water. If one watered-out zone is choked back while the other zones are still open and predominantly produce oil, then the only way to increase the oil production from the well is by "pulling the well harder", i.e. by opening the wellhead choke more and thereby reduce the bottomhole pressure. To what extent this is practically possible, depends on the well's operational constraints.

Operational Constraints

If the well or field is limited by liquid-handling capacity, a reduction in water rate caused by the AICDs will allow the oil rate to be increased by the same amount, such that the total liquid production (oil + water) is still the same. If the well or field is limited by water-handling capacity, the increase in oil rate can in principle be much larger than the reduction in water rate, but this scenario is less likely.

Many other operational constraints may also apply to a well, like:

- Gas-handling capacity
- Erosion
- Reservoir pressure

- Separator pressure
- Flowline inlet pressure, which may be affected by line-packing to prevent riser slugging
- Maximum reservoir drawdown due to weak formations and risk of borehole collapse
- Minimum ESP intake pressure
- Minimum casing shoe pressure
- Drainage strategy
- Regulatory restrictions on CO₂, H₂S, flaring, etc.

Quantifying the potential of an AICD completion to reduce the well's overall water cut and cumulative water production, and at the same time increase the oil rate and frontload the oil-production profile, can sometimes be challenging because it depends on the active operational constraint, which may change several times throughout the well's life. This—combined with reservoir uncertainty—makes it desirable to have AICDs with wide operational envelope and robustness against unforeseen events, such that the risk of unintended, excessive oil choking is minimized.

Consequences of Increased Pressure Drop

Viscosity-based and density-based technologies may respond differently to an increase in pressure drop across the lower completion. This is illustrated in Fig. 6, which is the same as Fig. 3, but overlaid with additional information and measured data (flow rate multiplied by eight). If the well initially produces no water and operates at the design point, the viscosity-based and density-based technologies will give the same oil rate and pressure drop. If some zones start to produce water and are choked back while the well is pulled harder to produce more oil out of zones that are still open and produce oil only, then the pressure drop and oil rate will increase for the latter zones.

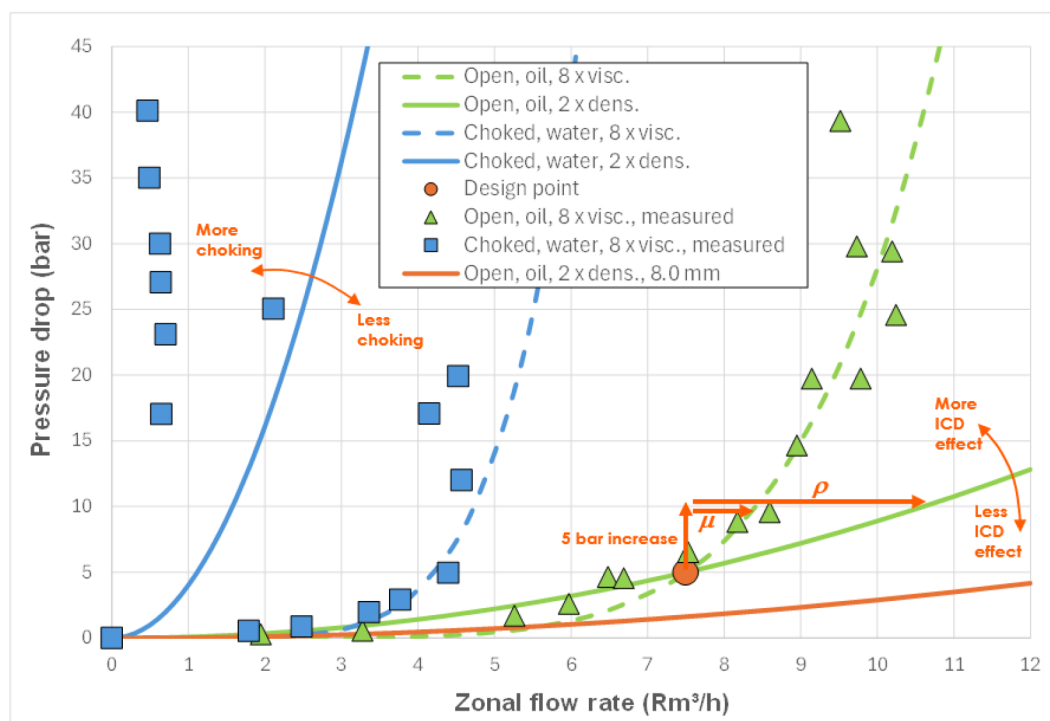


Figure 6—Zonal single-phase performance curves and effect of increased pressure drop.

However, as illustrated in Fig. 6, the potential for increased oil production depends on the slope of the performance curve around the operational point. One advantage of density-based technologies is that the performance curves are quadratic without the above-mentioned "hockey stick". This may allow significantly more oil to be produced for a moderate increase in pressure drop across the completion (reduction in bottomhole pressure).

Furthermore, density-based technologies have customizable performance curves, both in open and closed position, as indicated by the double-headed arrows in Fig. 6. By reducing the nozzle diameter in open position, the pressure drop will increase, which promotes a more uniform inflow distribution from the reservoir, or "ICD effect". On the contrary, if the objective is to minimize the pressure drop in open position, a larger nozzle diameter should be selected. Fig. 6 includes an example (orange curve) with two density-based devices which each have 8.0 mm nozzle.

Even if density-based technologies have higher flow capacity than viscosity-based technologies, there are still some operational limitations. For the DAR Technology that was used in the case study above, the working principle relies on the incoming flow being stratified. At liquid rates higher than about 6 Rm³/h (i.e. 12 Rm³/h for 2 × DAR), onset of droplet formation at the oil-water interface may cause the DAR Assembly to close at a water fraction that increases slightly with flow rate, but the operational consequences of this will normally be limited.

Fig. 6 also shows the performance curves for single-phase water. At low and moderate pressure drops, the density-based technology can give higher choking efficiency, as expected for an ultra-light oil, but it can also be seen from the measured data how the viscosity-based technology gives higher choking efficiency at high pressure drops as well as a hysteresis effect, which are unfortunately not captured by the extended RCP function in Eq. 1.

Conclusions

1. In this case study, it was shown how two density-based AICDs can provide the same oil-flow capacity as eight viscosity-based AICDs and also provide higher water-choking efficiency.
2. In order to account for fluid separation in the annulus and potential cascade closing of AICDs, it is recommended to use the workflow proposed by Langaas et al. (2024).
3. Different AICD technologies can exhibit very different two-phase behaviour, the importance of which increases with decreasing number of AICDs in a zone.
4. The binary performance of density-based AICDs and ability to remain fully open until higher water fractions are reached, can reduce the risk of unintended premature cascade closing and excessive choking of oil production.
5. For each AICD that closes in a zone, the relative reduction in total zonal flow area increases. For density-based AICDs, this means that the last device in a zone should ideally not close until the zonal water fraction has reached the level where maximum choking is desirable.
6. If AICDs choke back water in one zone and the well is pulled harder to produce more oil from the other zones, then the density-based technologies will typically have higher potential for increased oil production because the performance curve for oil is quadratic without "hockey stick" shape.
7. In general, it is important for AICDs to have a wide operational envelope and sufficient design flexibility to allow for reservoir uncertainty, changing operational conditions and constraints, and robustness against unforeseen events.

Nomenclature

- a, b, c, d, e, f = tuning parameters in extended RCP function (-)
 d = diameter (mm)
 E = choking efficiency (-)

p = pressure (bar)
 q = flow rate (m³/d)
 s = strength parameter (bar·m³/kg)
 x = tuning parameter (-)
 y = tuning parameter (-)
 z = tuning parameter (-)
 α = fluid phase fraction (-)
 μ = fluid viscosity (cP)
 ρ = fluid density (kg/m³)

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